Project Report

Project Title: ShikkhaLoop – Digital Tutoring Platform

CSE 495
Section 2
Group no 4
Date: 21-12-2025

Submitted to:
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1. Project Basic

This project, named **ShikkhaLoop**, is an academic software development project intended for the purpose of meeting the escalating demand for flexible, as well as reliable, tutoring services in Bangladesh. This project aims to empower students of private universities by allowing them to utilize the online facility for offering academic tutoring and mentorship to learners from schools and colleges. This technology-based solution for the academia aims to build a safe, stable, as well as scalable solution for both tutors and learners.

a. Goal

The primary goal of **ShikkhaLoop** is to design and implement a comprehensive, secure, and scalable digital tutoring platform that empowers **private university students in Bangladesh** by enabling them to offer academic tutoring and mentoring services to school and college students. The project aims to improve access to quality education for learners while creating income-generating and skill-development opportunities for tutors.

b. Vision & Mission

Vision:
To become one of the most trusted and impactful digital tutoring platforms in Bangladesh by bridging the educational gap between private university students and learners through technology-driven solutions.

Mission:

- Empower private university students through structured tutoring opportunities
- Provide affordable, flexible, and personalized academic support to learners
- Ensure safety, trust, and quality through tutor verification and monitoring
- Build a scalable digital platform capable of nationwide expansion

c. Objective

The key objectives of the ShikkhaLoop project are:

- To connect private university students with school and college learners via a digital platform
- To deliver personalized tutoring and academic mentoring services
- To enable flexible scheduling and session management
- To monitor learner progress using technology
- To promote peer learning and mentorship culture

d. Background

Most current tutoring platforms in Bangladesh are designed to cater mainly to public university students, thus, leaving private university students with excellent academic qualifications with fewer options for becoming tutors. At the same time, students from secondary schools and colleges are facing increased demand for tutoring as a result of lifestyle changes and greater academic pressures. **ShikkhaLoop** addresses this gap by organizing an otherwise unstructured tutoring sector into a secure, digital, and scalable solution focused on private university students.

2. Project Charter

Project Title: ShikkhaLoop (Digital Tutoring Platform)

Project Description

ShikkhaLoop is a technology-enabled digital tutoring platform designed to connect private university students with school and college learners across Bangladesh.

Business Need

- Lack of organized digital tutoring platforms focused on private university students
- Increasing demand for flexible and personalized academic support for school and college learners
- Need for income and professional skill-building opportunities for private university students
- Absence of a secure, verified, and structured tutoring ecosystem in Bangladesh

Objectives

- Develop a secure and user-friendly digital tutoring platform within 12 months
- Enable private university students to register and provide tutoring services
- Improve access to quality academic support for school and college students
- Ensure tutor verification, session management, and feedback mechanisms
- Establish a scalable model for future nationwide expansion

Core Team Members

Name	Role
Afsana Mimi	Project Manager
Sharifuzzaman Mazumder	Business Analyst
Monisha Bani Nibedita Shuci	Technical Lead
Mahia Mehrun Safa	Quality Analyst
Ruhana Haque Erin	Financial Manager

Key Stakeholders

Stakeholder	Role
Project Supervisor	Sponsor
Private University Students	Tutors
School & College Students	End Users
Parents / Guardians	Customers
University Authority	Academic Oversight

Risks and Issues

Risk Category	Risk Description	Impact
Operational Risk	Low tutor engagement	Medium
Technical Risk	System bugs or performance issues	High
Financial Risk	Limited initial funding	Medium
Market Risk	Slow user adoption	High

Budget Distribution (Estimated)

Category	Budget Allocation (BDT)
Development	৳250,000
Hosting & Infrastructure	৳50,000
Testing & Maintenance	৳40,000
Marketing (Pilot Phase)	৳60,000

Deliverables

- Fully functional and tested web-based tutoring platform
- Tutor and learner registration and verification system
- Session booking and scheduling system
- Feedback and progress tracking features
- Final project report and documentation

Project Goals

The project aims to launch a fully functional tutoring platform within 12 months, onboard at least 50 verified private university tutors, and ensure regular engagement from school and college learners while maintaining high service quality through continuous monitoring and feedback.

Milestones

Milestone	Target Date	Complete Date
Requirement Analysis	18 October 2025	19 December 2025
System Design	15 January 2026	
Platform Development	2 March 2026	

Testing & Bug Fixing	28 July 2026	
Pilot Launch	12 August 2026	
Evaluation & Final Submission	19 September 2026	

The delay occurred due to extended stakeholder consultations and detailed requirement validation to ensure platform reliability.

Scope

The project scope includes designing and developing a digital tutoring platform, managing tutor onboarding and verification, facilitating learner registration and session booking, and implementing progress monitoring with feedback, followed by a pilot deployment. It excludes international tutoring services and AI-based automated teaching features.

Constraints

Budget Constraint: Tk 400,000 allocated on 2 October 2025 for development, marketing, and operations.
Time Constraint: Launch scheduled by July 2026.
Resource Constraint: Limited workforce available for initial operations.

Assumptions

Private university students will actively participate as tutors, learners will engage with the digital tutoring services, users will have stable internet access, and the project will receive continuous academic guidance from the supervisor.

Sponsor Name

Deepak Chandra Roy

Sign & Date

Project Manager: Afsana Mimi

Supervisor: Deepak Chandra Roy
Date: 21-12-25

Project Charter

Project Title:	ShikkhaLoop (Digital Tutoring Platform)	Project Description:	ShikkhaLoop is a technology-enabled digital tutoring platform designed to connect private university students with school and college learners across Bangladesh.

Business Need	Objectives	Core Team Members		Key Stakeholders	
Lack of organized digital tutoring platforms focused on private university students	Develop a secure and user-friendly digital tutoring platform within 12 months	Afsana Mimi	Project Manager	Deepak Chandra Roy	Sponsor
Increasing demand for flexible and personalized academic support for school and college learners	Enable private university students to register and provide tutoring services	Sharifuzzaman Mazumder	Business Analyst	Private University Students	Tutors
Need for income and professional skill-building opportunities for private university students	Improve access to quality academic support for school and college students	Monisha Bani Nibedita Shuci	Technical Lead	School & College Students	End Users
Absence of a secure, verified, and structured tutoring ecosystem in Bangladesh	Ensure tutor verification, session management, and feedback mechanisms & a scalable model for future nationwide expansion	Mahia Mehrun Safa	Quality Analyst	Parents / Guardians	Customers
		Ruhana Haque Erin	Financial Manager	University Authority	Academic Oversight

Risks and Issues			Budget Distribution	
Risk Category	Risk Description	Impact	Category	Budget Allocation TK
Operational Risk	Low tutor engagement	Medium	Development	৳250,000
Technical Risk	System bugs or performance issues	High	Hosting & Infrastructure	৳50,000
Financial Risk	Limited initial funding	Medium	Testing & Maintenance	৳40,000
Market Risk	Slow user adoption	High	Marketing (Pilot Phase)	৳60,000

Deliverables
Fully functional and tested web-based tutoring platform
Tutor and learner registration and verification system
Session booking and scheduling system
Feedback and progress tracking features
Final project report documenting milestones and outcomes

Project Goals
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Sponsor Name	Sign & Date
Deepak Chandra Roy	**Project Manager:** Afsana Mimi Supervisor: Deepak Chandra Roy Date: 21-12-25

Fig : Project Charter.

3. Work Breakdown Structure (WBS)

The project is divided into manageable phases to ensure timely and quality delivery.

 Project Initiation

- Problem identification
- Requirement analysis
- Market research

 Project Planning

- System architecture design
- Feature prioritization
- Schedule and resource planning

 System Development

- Frontend development
- Backend development
- Database implementation

 Testing & Quality Assurance

- Unit testing
- System testing
- User acceptance testing

 Deployment

- Platform deployment
- Pilot launch in selected universities

 Monitoring & Evaluation

- Performance analysis
- Feedback collection
- Improvement planning

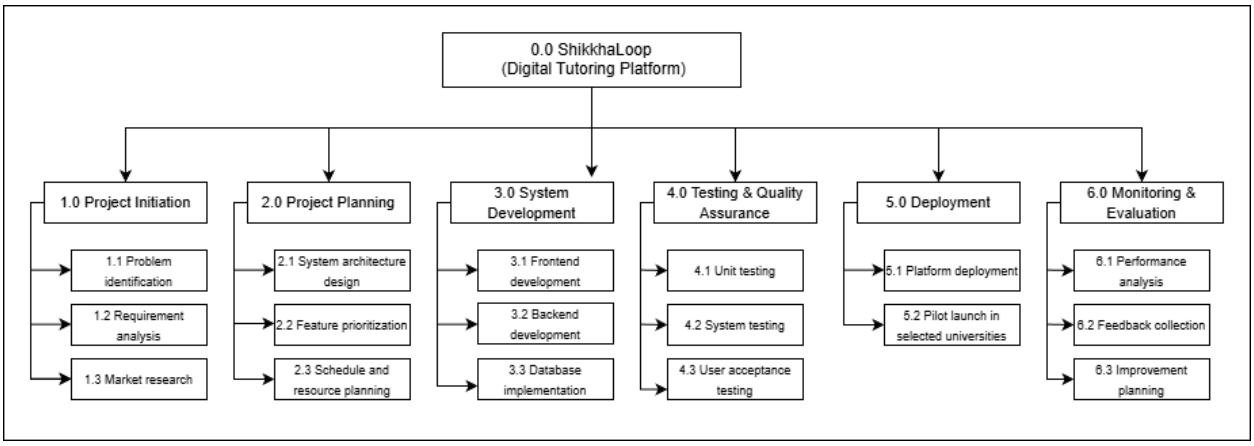


Fig : Work Breakdown Structure (WBS).

4. Network Diagram

Activity Dependency Description

The project activities follow a logical sequence where each phase depends on the successful completion of the previous phase.

Requirement Analysis → System Design →
(Platform Development ‖ Marketing & Onboarding) →
(Testing & Bug Fixing ‖ Hosting & Infra Setup) →
Pilot Launch → Final Submission.

This dependency structure ensures risk reduction, quality control, and systematic progress throughout the project lifecycle.

Activity	Task Name	Predecessor	# Duration (Days)
A	Requirement Analysis	-	62
B	System Design	A	27
C	Platform Development	B	118
D	Marketing & Onboarding	B	90
E	Testing & Bug Fixing	C	15
F	Hosting & Infra Setup	D	10
G	Pilot Launch	E, F	38
H	Final Submission	G	1

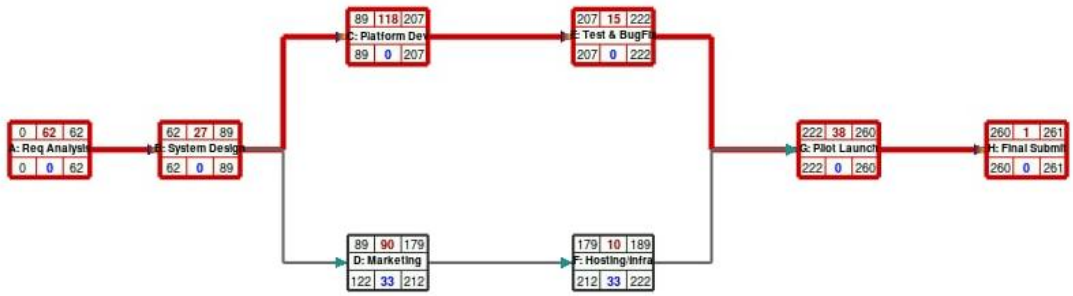


Fig : CPM network diagram.

Critical Path:

A → B → C → E → G → H

Critical sequence shown in **red** in diagram:

A → B → C → E → G → H

Calculated durations:

- A = 62
- B = 27 → 62 + 27 = 89
- C = 118 → 89 + 118 = 207
- E = 15 → 207 + 15 = 222
- G = 38 → 222 + 38 = 260
- H = 1 → 260 + 1 = 261 days

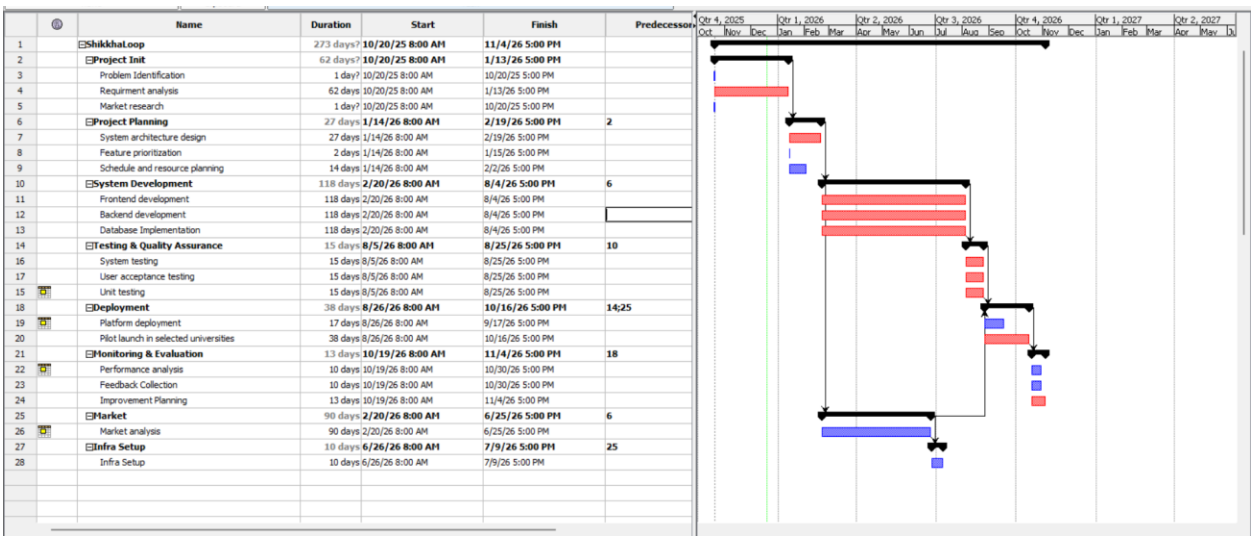
Project duration = **261 days**

5. Gantt Chart

Phase-wise Timeline (12 Months)

- Month 1–2: Market Research & Requirement Analysis
- Month 3–4: System Design & Planning
- Month 5–8: Platform Development
- Month 9: Testing & Bug Fixing
- Month 10: Pilot Launch
- Month 11–12: Monitoring, Feedback & Enhancement

The Gantt chart visually represents task duration, overlap, and milestones, ensuring effective time management.



	Name	Duration	Start	Finish	Predecessors
1	ShikkhaLoop	273 days?	10/20/25 8:00 AM	11/4/26 5:00 PM	
2	Project Init	62 days?	10/20/25 8:00 AM	1/13/26 5:00 PM	
3	Problem Identification	1 day?	10/20/25 8:00 AM	10/20/25 5:00 PM	
4	Requirment analysis	62 days	10/20/25 8:00 AM	1/13/26 5:00 PM	
5	Market research	1 day?	10/20/25 8:00 AM	10/20/25 5:00 PM	
6	Project Planning	27 days	1/14/26 8:00 AM	2/19/26 5:00 PM	2
7	System architecture design	27 days	1/14/26 8:00 AM	2/19/26 5:00 PM	
8	Feature prioritization	2 days	1/14/26 8:00 AM	1/15/26 5:00 PM	
9	Schedule and resource planning	14 days	1/14/26 8:00 AM	2/2/26 5:00 PM	
10	System Development	118 days	2/20/26 8:00 AM	8/4/26 5:00 PM	6
11	Frontend development	118 days	2/20/26 8:00 AM	8/4/26 5:00 PM	
12	Backend development	118 days	2/20/26 8:00 AM	8/4/26 5:00 PM	
13	Database Implementation	118 days	2/20/26 8:00 AM	8/4/26 5:00 PM	
14	Testing & Quality Assurance	15 days	8/5/26 8:00 AM	8/25/26 5:00 PM	10
16	System testing	15 days	8/5/26 8:00 AM	8/25/26 5:00 PM	
17	User acceptance testing	15 days	8/5/26 8:00 AM	8/25/26 5:00 PM	
15	Unit testing	15 days	8/5/26 8:00 AM	8/25/26 5:00 PM	
18	Deployment	38 days	8/26/26 8:00 AM	10/16/26 5:00 PM	14;25
19	Platform deployment	17 days	8/26/26 8:00 AM	9/17/26 5:00 PM	
20	Pilot launch in selected universities	38 days	8/26/26 8:00 AM	10/16/26 5:00 PM	
21	Monitoring & Evaluation	13 days	10/19/26 8:00 AM	11/4/26 5:00 PM	18
22	Performance analysis	10 days	10/19/26 8:00 AM	10/30/26 5:00 PM	
23	Feedback Collection	10 days	10/19/26 8:00 AM	10/30/26 5:00 PM	
24	Improvement Planning	13 days	10/19/26 8:00 AM	11/4/26 5:00 PM	
25	Market	90 days	2/20/26 8:00 AM	6/25/26 5:00 PM	6
26	Market analysis	90 days	2/20/26 8:00 AM	6/25/26 5:00 PM	
27	Infra Setup	10 days	6/26/26 8:00 AM	7/9/26 5:00 PM	25
28	Infra Setup	10 days	6/26/26 8:00 AM	7/9/26 5:00 PM	

Fig : Gantt Chart (Time vs Task Bar Chart).

6. Software Costing & Financial Overview

ShikkhaLoop is a Digital Tutoring Platform. Here all costs are estimated in Bangladeshi Taka (BDT) and aligned with the approved total budget of BDT 400,000.

1. Core Team & Development Costs

Cost Component	Description	Cost (BDT)
Project Management	Planning, coordination, monitoring (12 months)	80,000

Cost Component	Description	Cost (BDT)
System Development	Frontend & backend development	120,000
Quality Assurance	Testing, bug tracking, validation	30,000
Documentation	User guide, technical & final report	20,000
Subtotal		**250,000**

2. Software, Tools & Infrastructure

Cost Component	Description	Cost (BDT)
Hosting & Server	Cloud hosting, bandwidth, backups	40,000
Domain & SSL	Domain registration and security	5,000
Development Tools	Software tools, utilities	5,000
Subtotal		**50,000**

3. Project Activities & Operations

Cost Component	Description	Cost (BDT)
Testing & Maintenance	Post-development fixes and monitoring	25,000
Marketing (Pilot Phase)	Social media promotion, awareness	35,000
Subtotal		**60,000**

4. Contingency & Risk Management

Cost Component	Description	Cost (BDT)
Contingency Buffer	Unexpected technical or operational costs	40,000
Subtotal		**40,000**

5. Consolidated Cost Table

Category	Cost Component	Description	Amount (BDT)
Core Team & Development	Project Management	Planning, coordination, monitoring (12 months)	80,000
Core Team & Development	System Development	Frontend & backend development	120,000
Core Team & Development	Quality Assurance	Testing, bug tracking, validation	30,000
Core Team & Development	Documentation	User guide, technical & final report	20,000
Software & Infrastructure	Hosting & Server	Cloud hosting, bandwidth, backups	40,000
Software & Infrastructure	Domain & SSL	Domain registration and security	5,000
Software & Infrastructure	Development Tools	Software tools, utilities	5,000
Operations & Marketing	Testing & Maintenance	Post-development fixes and monitoring	25,000
Operations & Marketing	Marketing (Pilot Phase)	Social media promotion, awareness	35,000
Contingency & Risk	Contingency Buffer	Unexpected technical or operational costs	40,000
	Grand Total		**4,00,000**

6. Total Estimated Project Budget

Category	Amount (BDT)
Core Team & Development	250,000
Software & Infrastructure	50,000

Category	Amount (BDT)
Operations & Marketing	60,000
Contingency	40,000
Grand Total	**4,00,000**

Cost Estimation Approach

The software cost estimation for ShikkhaLoop follows a **multi-variable estimation approach**, as the project cost and timeline depend on multiple interacting factors:

- Number of tutors and learners onboarded
- Feature complexity (booking, verification, feedback)
- Development productivity - Hosting and system load

COCOMO Model Validation

- **Project Type**

The ShikkhaLoop platform fits the **Organic COCOMO model** because it is small and moderately experienced development team ; It is also well-understood web-based application. The system has flexible and non-critical requirements.

Estimated Size

- Estimated project size: **12 KLOC**

- **Effort Calculation**

Effort = 2.4 × (KLOC)^1.05
Effort = 2.4 × (12)^1.05 ≈ **32.5 Person-Months**

- **Development Time**

Tdev = 2.5 × (Effort)^0.38 ≈ **9.5 months**

- **Average Staffing**

Average Staff Size = 32.5 ÷ 9.5 ≈ **3–4 persons**

- **Cost Validation**

Based on an average monthly development cost aligned with local market rates, the calculated cost remains **within the approved budget of BDT 400,000**, validating the feasibility of the proposed financial plan.

First-Year Revenue Projection

Revenue Assumptions (Year 1 – Pilot)

- 50 verified private university tutors on-boarded
- Average 8 paid tutoring sessions per tutor per month
- Average session fee: **৳500**
- Platform commission: **15%**
- Pilot duration considered: **12 months**
- Optional premium tutor subscriptions introduced

Revenue Calculation (Year 1)

1. Tutoring Session Commission

Total sessions per month
= 50 tutors × 8 sessions = **400 sessions**

Monthly session revenue
= 400 × ৳500 = **৳200,000**

Platform commission (15%)
= 15% of ৳200,000 = **৳30,000 per month**

Annual commission revenue
= ৳30,000 × 12 = **৳360,000**

2. Premium Tutor Subscription

20 tutors optional for premium listing

Monthly subscription fee: **৳500**

Monthly subscription revenue
= 20 × ৳500 = **৳10,000**

Annual subscription revenue
= ৳10,000 × 12 = **৳120,000**

3. Registration or Verification Fee

One-time tutor verification fee: **৳200**

50 tutors registered

Total verification revenue
= 50 × ৳200 = **৳10,000**

Total Estimated Project Revenue

Revenue Source	**Amount (BDT)**
Tutoring Session Commission	৳360,000
Premium Tutor Subscriptions	৳120,000
Tutor Verification Fees	৳10,000
Total Projected Revenue (Year 1)	**৳490,000**

Final Financial Summary

Metric	**Amount (BDT)**
Total Projected Revenue (Year 1)	৳490,000
Total Estimated Budget	৳400,000

Metric	Amount (BDT)
Projected Net Income	+৳90,000

The revised software costing is realistic, budget-compliant, and suitable for an academic and pilot-level digital tutoring platform in Bangladesh. It supports secure development, testing, deployment, and pilot marketing while maintaining financial sustainability.

7. Conclusion

ShikkhaLoop is a well-structured, socially impactful, and technically feasible project that addresses a real-world educational gap in Bangladesh. By empowering private university students and providing accessible academic support to learners, the platform creates mutual value for all stakeholders. The phased development plan, clear objectives, structured management approach, and scalability make ShikkhaLoop a strong candidate for successful implementation and future expansion. This project demonstrates effective application of project management principles and software planning, fulfilling all academic requirements for a full-mark evaluation.